

$$X_0 \sim P_{\text{init}}$$

$$g(X_0) \sim \underbrace{P_{\text{data}}}_{Z_1 \dots Z_n}$$

$$\text{design } g \text{ s.t.} \\ g(X_0) \sim P_{\text{data}}$$

$$\frac{dX_t}{dt} = u_t(X_t)$$

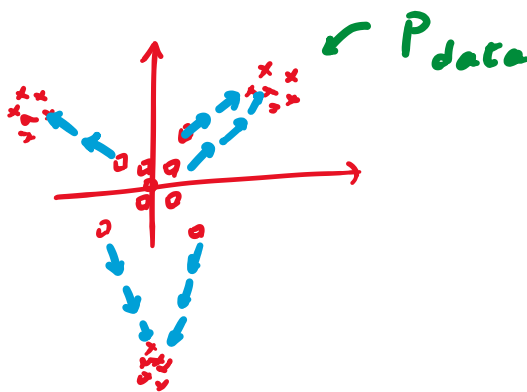
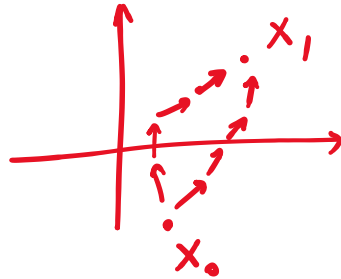
$$X_0 \sim P_{\text{init}}$$

$$X_1 \sim P_{\text{data}}$$

$$\text{design } u_t \dots$$

$$\forall 0 \leq t \leq 1$$

$$X_t \in \mathbb{R}^2$$



$$\text{Goal: Find } u_t^{\text{target}}$$

$$\text{s.t. when used in}$$

$$\text{ODE } \frac{dX_t}{dt} = u_t^{\text{target}}(X_t)$$

$$\text{results in } X_1 \sim P_{\text{data}}$$

$$\min_{\theta} \mathbb{E} \left\| u_{\theta}^t(x) - \underbrace{u_t^{\text{target}}(x)} \right\|_2^2$$

① Prob. Path

$$Z \sim P_{\text{data}}$$

$$X_t | Z \sim P_t(\cdot | \underline{z})$$

$$\leadsto P_0(\cdot | Z) = P_{\text{init}}(\cdot), \quad P_1(\cdot | Z) = \delta(\cdot - Z)$$

$$\frac{dX_t}{dt} = \underbrace{u_t(X_t|Z)}_{\text{define s.t. } X_t \sim P_t(\cdot|Z)} \quad . \leq t \leq 1$$

Example : $P_{\text{init}} = \mathcal{N}(0, I)$.

$$P_t(x|Z) = \mathcal{N}(\alpha_t Z, \beta_t^2 I)$$

$$\begin{cases} \alpha_0 = 0 & \beta_0 = 1 \\ \alpha_1 = 1 & \beta_1 = 0 \end{cases} \quad P_0(x|Z) = \mathcal{N}(0, I)$$

$$u_t(x|Z) = ? \quad X_t|Z = \alpha_t Z + \beta_t x \quad x \sim \mathcal{N}(0, I)$$

$$x = \frac{x' - \alpha_t Z}{\beta_t}$$

$$\frac{dX_t}{dt} = u_t(\underbrace{\alpha_t Z + \beta_t x}_{x'})|Z = \dot{\alpha}_t Z + \dot{\beta}_t x$$

$$\underbrace{u_t(x'|Z)} = \dot{\alpha}_t Z + \dot{\beta}_t \left(\frac{x' - \alpha_t Z}{\beta_t} \right)$$

$$= \left(\dot{\alpha}_t - \frac{\dot{\beta}_t \alpha_t}{\beta_t} \right) Z + \frac{\dot{\beta}_t}{\beta_t} x'$$

Q : How does $u_t(x|Z)$ & $u_t(x)$ relate to each other?

Continuity Lemma:

$$X_0 \sim P_{\text{init}} \quad \frac{dX_t}{dt} = u_t(X_t) \quad 0 \leq t \leq 1$$

$$X_t \sim P_t \quad \Leftrightarrow \quad \frac{\partial}{\partial t} P_t(x) = -\operatorname{div}(\underbrace{P_t \cdot u_t}(x)).$$

Divergence

$$\nabla \cdot F \quad \operatorname{div}(F)$$

$$F: \mathbb{R}^d \rightarrow \mathbb{R}^d$$

$$\nabla \cdot F := \sum_{i=1}^d \frac{\partial F_i}{\partial x_i}$$

$$\nabla \cdot (\underbrace{f}_{\in \mathbb{R}} \cdot \underbrace{F}_{\in \mathbb{R}^d}) = \nabla f \cdot F + f \nabla \cdot F$$

$$\hookrightarrow \text{Lemma:} \quad \int_{\mathbb{R}^d} \widetilde{\varphi(x)} \cdot f_1(x) dx = \int_{\mathbb{R}^d} \varphi(x) \cdot f_2(x) dx$$

$$\Leftrightarrow \quad f_1 \equiv f_2 \quad \frac{\partial}{\partial t} \Phi_t(x) = u_t(\Phi_t(x)).$$

$$\forall \varphi \in C_c^\infty: \underbrace{\int \varphi(x) \cdot P_t(x) dx}_{(1)} = \underbrace{\int \varphi(\Phi_t(x)) \cdot P_0(x) dx}_{(2)}$$

$$(1) \rightarrow \frac{d}{dt} \int \varphi(x) \cdot P_t(x) dx = \int \frac{\partial}{\partial t} P_t(x) \cdot \varphi(x) dx$$

$$\textcircled{2} \quad \frac{d}{dt} \int \varphi(\Phi_t(x)) \cdot p_t(x) dx = \int \nabla \varphi(\Phi_t(x)) \cdot \underbrace{\frac{\partial \Phi_t(x)}{\partial t} \cdot p_t(x)}_{u_t(\Phi_t(x))} dx$$

$$= \int \nabla \varphi(\Phi_t(x)) \cdot u_t(\Phi_t(x)) \cdot \underbrace{p_t(x)}_{p_t(x)} dx$$

$$= \int \nabla \varphi(x) \cdot u_t(x) \cdot p_t(x) dx$$

$$\forall \varphi \in C_c^\infty \quad \int \frac{\partial}{\partial t} p_t(x) \cdot \varphi(x) dx = \int \underbrace{\nabla \varphi(x)}_f \cdot \underbrace{u_t(x) \cdot p_t(x)}_F dx$$

$$\nabla \cdot (f \cdot F) = \underbrace{\nabla f \cdot F}_{} + f \cdot (\nabla \cdot F)$$

$$= \int \left[\nabla \cdot (\varphi(x) \cdot u_t(x) \cdot p_t(x)) - \varphi(x) \cdot \text{div}(p_t u_t) \right] dx$$

$$= \underbrace{\int \nabla \cdot (\varphi(x) \cdot u_t(x) \cdot p_t(x)) dx}_{\oint_{\text{Surf}(\varphi)} \tilde{r}_t(x) \cdot \vec{n} dx = 0} - \int \varphi(x) \text{div}(p_t u_t) dx$$

$$\oint_{\text{Surf}(\varphi)} \tilde{r}_t(x) \cdot \vec{n} dx = 0$$



$$\Rightarrow \forall \varphi \in C_c^\infty : \int \frac{\partial}{\partial t} P_t(x) \varphi(x) dx$$

$$= \int -\operatorname{div}(P_t u_t)(x) \varphi(x) dx$$

$$\Rightarrow \frac{\partial}{\partial t} P_t(x) = -\operatorname{div}(P_t u_t)(x) \quad \square$$

Theorem: $\underbrace{u_t(x)} = \int \underbrace{u_t(x|z)} \cdot \frac{P(x|z) P_{\text{data}}(z)}{P_t(x)} dz.$

Proof: $P_t(x) = \int P_t(x|z) P_{\text{data}}(z) dz$

$$\rightarrow \frac{\partial}{\partial t} P_t(x) = \int \frac{\partial}{\partial t} P_t(x|z) \cdot P_{\text{data}}(z) dz$$

$$= \int -\operatorname{div}(u_t(\cdot|z) \cdot P_t(\cdot|z))(x) \cdot P_{\text{data}}(z) dz$$

$$= -\operatorname{div} \left[\int \underbrace{u_t(x|z) P_t(x|z) \cdot P_{\text{data}}(z)}_{P_t(x)} \cdot \underbrace{V_t(u)}_{\substack{\text{vector} \\ \text{field}}} dz \right]$$

$$= -\operatorname{div}(P_t(x) \underbrace{V_t(x)}_{\substack{\text{vector} \\ \text{field}}})$$

$\Rightarrow u_t(x)$ that makes $P_t(x)$ is exactly

$$\int u_t(x|z) \cdot \frac{P_t(x|z) P_{\text{data}}(z)}{P_t(x)} dz. \quad \square$$